

Experimental investigation of reforming and flow characteristics of a steam methane reformer filled with nickel catalyst of various shapes

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ARTICLE INFO

Keywords:

Methane reforming
Catalyst
Packed bed
Methane conversion
Nickel catalyst
Catalyst shape

ABSTRACT

An experimental investigation of the steam methane reforming process in a cylindrical packed bed filled with the $\text{Ni-}\alpha\text{-Al}_2\text{O}_3$ catalysts was performed to understand the effect of particle shapes of catalyst on a pressure drop and a methane conversion. Four particle shapes such as a simple cylinder, a Raschig ring, a 7-holes cylinder, and a 7-holes sphere were tested. Moreover, the simple cylinders and the Raschig rings were tested for various end-to-end dimensions (5 mm, 10 mm, 15 mm, 20 mm, 25 mm). The present study showed that the methane conversion is increasing with an increase in the geometric surface of the packed at a constant packed bed length (600 mm). Two quantities are proposed for evaluating the effectiveness of the packed beds filled with a catalyst of various shape: a weight residence time ($\text{kg}_{\text{cat}}\cdot\text{s}/\text{mol}$) and a surface residence time ($\text{m}^2\cdot\text{s}/\text{mol}$). The maximum methane conversion is observed for the 7-holes spheres. The catalytic activity of the surface unit of all investigated catalyst shapes is approximately the same. A characteristic quadratic dependence of a pressure drop versus velocity $\Delta P = f(u^2)$ is observed for all catalyst types. The minimum pressure drops are taking place in a packed bed filled with the 7-holes cylinders, while the maximum pressure drops are occurring for the simple cylinders. The linear ($1/\text{mm}$) and volumetric ($1/\text{m}^3$) drag coefficient were calculated. The drag coefficient is decreasing when the velocity decreases, especially in the velocity range up to 1 m/s.

1. Introduction

The reforming processes of hydrocarbons on the solid catalysts is widespread in the petrochemical industry for the hydrogen (H_2) and synthesis gas (syngas) production. Among the oxidizing agents for reforming, the most common are steam and carbon dioxide. It is known that steam reforming of natural gas is the main process for the production of hydrogen [1–3] and synthesis gas [4–6]. About 90% of hydrogen in the world industry is produced from fossil fuel sources, for example from natural gas, which is mainly composed of methane. The steam methane reforming (SMR) process is one of the most common hydrogen production technology from natural gas. Steam methane reforming accounts for about half of the global production of hydrogen [7]. In addition, steam methane reforming is becoming more common in the thermochemical waste-heat recuperation systems [8–11].

In the steam methane reforming process methane (CH_4) reacts with steam in the presence of a catalyst to produce hydrogen (H_2), carbon monoxide (CO), and carbon dioxide (CO_2). Various types of catalysts are used to increase the efficiency of this process. The nickel based catalysts are widely used in large-scale hydrogen industry [12–16]. Moreover, the noble metals such as ruthenium (Ru) [17], rhodium (Rh) [18], palladium (Pd) [19], iridium (Ir) [20] and platinum (Pt) [21] are

used as the active element in catalysts.

A steam methane reformer is the core element for steam methane reforming [7,22–24]. There are several typical configurations of methane reformers [25,24]. The tubular reformers are widespread for the SMR process [25–27]. In such reformers, the reaction space is filled with catalysts. As mentioned above, the nickel-based catalyst is most often used. Packed beds with random packing of catalysts particles are widely used for the tubular methane reformers [28,29]. Catalyst particles of different sizes and shapes are randomly placed in a packed bed of reformer. Reaction mixture flows over catalyst particles and reactions of steam methane reforming and heat transfer take place simultaneously on the external (particle scale) and internal (pore scale) surfaces of catalysts particles. The catalysts particles commonly have spherical or cylindrical shapes. However other shapes with different internal structures (voids) or external shapes could be used for the SMR process [28].

The SMR is a well-studied process, both theoretically and experimentally [30]. The study of the SMR process can be divided into two main research trend – experimental investigation and CFD-modeling. With the help of experimental investigation, data arrays can be obtained that can be used to develop and verify models for CFD-modeling [31–34]. Researchers and experts from different countries are mainly

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<https://doi.org/10.1016/j.enconman.2019.01.103>

Received 2 December 2018; Accepted 23 January 2019

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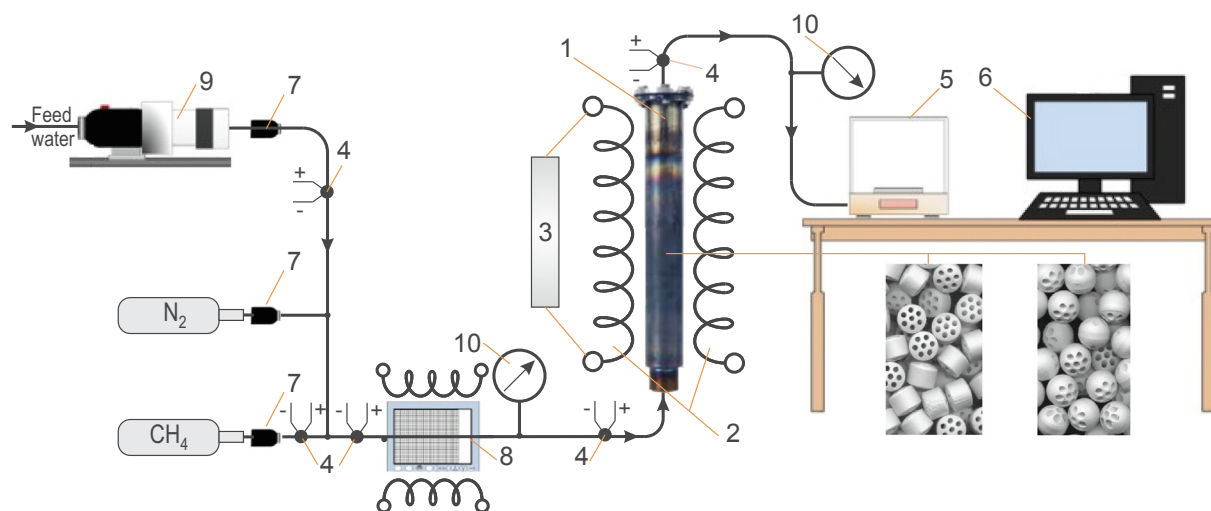


Fig. 1. Schematic view of experimental setup.

focusing on SMR investigations on the development of new catalysts, designs of reformers, the SMR process diagrams, etc. In the publications on SMR, it is possible to distinguish also two main directions – reforming characteristics and flow dynamics in a packed bed. Often, the results of SMR researches are difficult to compare due to the different properties of the catalyst and shape of catalyst particles, different technological parameters and different configurations of reformers, etc. Moreover, catalyst particle shapes will have an impact on the pressure drop of a packed bed, local fluid flow, and therefore heat transfer and reaction performance of a packed bed [35,36].

Commonly researchers study the characteristics of the SMR process for one shape of a catalyst. For example, comprehensive paper on the experimental investigations of SMR presented by Xu and Froment [12] is considering only the spherical catalyst particles. Moreover, among the experimental studies in which only spherical particles of the catalyst has been considered, several articles can be distinguished. Hoang et al. [37] reported results of a kinetic and modeling study of methane steam reforming over sulfide nickel catalyst on a gamma alumina support. The focal point of this article is on the reforming characteristics versus operation parameters such as temperature, steam-to-methane ratio, residence time, etc. The article presented by Hou and Hughes [38] is showing the results of experimental investigation of methane steam reforming over a commercial Ni/ α -Al₂O₃ catalyst in an integral reactor under conditions of no diffusion limitation. The essential results of this work are the dependences of methane reforming characteristics versus pressure in a tubular reformer. In addition, the results of experimental investigation of the SMR process for spherical or cylindrical shapes are reported in the other literary sources.

Another set of articles is devoted to the study of flow dynamics in a packed bed of methane reformers. In these papers, the focus is on studying pressure loss for various conditions. There is a number of papers that are devoted to an experimental investigation of pressure drop in the packed beds. Bai et al. [29] experimentally investigated a pressure drop in fixed bed reformer filled with randomly packed catalyst particles. They considered two catalyst shapes – cylindrical and spherical. Also, Calis and co-workers [39] investigated a pressure drop in a novel structured catalytic reactor packing. And then obtained experimental results were used for verification of numerical model of flow dynamics. In the indicated works of Calis and Bai does not have any results of the reforming characteristics. There is no literature (according to author's literature review) on combined experimental investigation of the reforming characteristics and pressure drop in the SMR process in terms of further effects of temperature, feed composition, catalyst shapes, etc.

There is a number of articles that are devoted to the combined

investigation of the reforming characteristics and pressure drop in the SMR process via CFD-modeling. Karthik and Buwa [28] reported the results of a numerical investigation of the effect of particle shape on fluid flow and heat transfer for methane steam reforming reactions in a packed bed. They considered six particle shapes: cylinder, trilobe and daisy having external shaping, hollow cylinder, cylcut and 7-hole cylinder including internal voids. The results of the article are of importance for the development of numerical models, but for engineering calculations, their results are difficult to use because the simulation was performed for a packed bed with only 9 particles. The real reformers have hundreds and thousands of catalyst particles.

It is well known that the efficiency of a methane reformers increases as the residence time (contact time) increases [38,37]. However, a higher residence time results in lower reformer output. In contrast, under very low residence time the velocity of the reaction mixture is increasing that leads to an increase of pressure drop and a decrease of methane conversion. Maximizing reforming efficiency with minimal pressure loss is an important engineering problem.

In the present work, the experimental approach is used to understand and to quantify the effect of particle shapes (cylinder, sphere, Raschig ring, cylinder with 7-holes) and operational parameters (such as temperature, inlet steam-to-methane ratio, residence time) on pressure drop and reaction characteristics. In addition, the local drag coefficients of a 1 meter and 1 m³ of a packed bed were calculated based on the experimental results.

2. Experiments

The equipment is classical. A schematic diagram of the experimental setup is presented in Fig. 1. The experimental setup can be divided into three blocks. The first block is the block for preparing of the reaction mixture. The second block is the block of the reforming process. The third block is the block for measuring and storing of the results. The main elements of the first block are an electrical steam generator and a preheater. The main element of the second block is a methane reformer. The main component of the third block is a portable gas analyzer.

2.1. Experimental setup

The experimental setup mainly consists of the following elements: 1 – a methane reformer; 2 – an electrical muffle furnace; 3 – a watt-hour meter; 4 – the thermocouples; 5 – a portable gas analyzer; 6 – a personal computer; 7 – the mass-flow meters; 8 – an electrical preheater; 9 – an electrical steam generator; 10 – a manometer.

The methane reformer is made of high-alloy steel and placed in an

electric muffle furnace. The tubular reformer is made of HK 40 stainless steel (25–20 Cr-Ni). The nickel of steel is not playing a role as a catalyst. The tube of reformer has an outer diameter of 10 cm and a length of 80 cm. However, the reaction zone containing the catalyst has a length of 60 cm. The reformer tube is heated by the electrical heaters. The electrical furnace heats the reaction space up to 500–600 °C. The thermocouples are made of a tungsten-rhenium alloy.

The gas composition is controlled by the portable infrared syngas analyzer Gasboard 3100P, which can be connected to a personal computer. High purity deionized distilled water is used for a steam production. Superheated steam is produced in the electric steam generator PG-LEK 1.1 (in Russian).

The methane and nitrogen were obtained from the cylinders. Nitrogen is used as neutral gas for catalyst preheating. The flows of methane are measured by rotameters and the flow of steam is controlled by the mass-flow meters. High purity methane N40 (99.99% purity) is used for experiments. Steam and methane are mixed and heated in an electrical heater. After the mixture heater, the homogeneous steam-methane mixture is passed into the steam methane reformer.

A photograph of the experimental setup and some elements of the schematic diagram are shown in Fig. 2. The numbers of the elements in Fig. 2 coincide with the figure caption to Fig. 1.

2.2. Catalyst

The reaction space of the methane reformer is filled with $\text{NiO-}\alpha\text{Al}_2\text{O}_3$ catalyst with the different catalyst particle shapes. For experiment the $\text{Ni-}\alpha\text{Al}_2\text{O}_3$ catalyst is used with the following chemical composition: $\text{NiO} = 14.5\%$; $\text{SiO}_2 = 0.2\%$; support $\text{CaO-MgO-La}_2\text{O}_3\text{-}\alpha\text{Al}_2\text{O}_3$ [40]. Other properties are presented below: bulk density is 680 kg/m^3 ; surface area is $4.5 \text{ m}^2/\text{g}_{\text{cat}}$; average porosity of catalyst particles is about 41%. The packed beds with structured packing of catalyst particles with the different shapes are presented in Fig. 3. There are four particle shapes: A – a simple cylinder; B – a Raschig ring (cylinder with one hole on axial line); C – a cylinder with 7 internal holes; D – a sphere with 7 internal holes.

The catalyst particles are placed into the tube with a diameter of 100 mm. The catalysts layer depth is 600 mm. The geometrical characteristics of catalysts are shown in Table 1.

Void fraction of packed bed (or interparticle porosity m^3/m^3) was

determined by two ways. First is as following:

$$\epsilon_{ip} = 1 - \frac{V_{\text{cat}}}{V_{rs}} = 1 - \frac{n_{\text{cat}} \cdot V_{\text{particle}}}{V_{rs}}, \quad (1)$$

where ϵ_{ip} – interparticle voidage; V_{cat} – volume of catalyst, m^3 ; V_{rs} – volume of reaction space of reformer, m^3 ; n_{cat} – number of catalyst particles in the reformer; V_{particle} – volume of one catalyst particle, m^3 .

Second is as following:

$$\epsilon_{ip} = 1 - \frac{W_{\text{cat}} \cdot \rho_{\text{cat}}}{V_{rs}}, \quad (2)$$

where W_{cat} – weight of catalyst, kg; ρ_{cat} – density of catalyst, kg/m^3 .

Scanning Electron Microscopy (SEM) image of the catalyst is presented in Fig. 3 (right side). SEM photo of catalyst structure and other catalyst properties in Table 1 is provided by the catalyst manufacturer NIAP-CATALYZATOR LLC (Novomoskovsk, Russia). The author thanks the manufacturer NIAP-CATALYZATOR LLC for the SEM image and catalyst properties.

2.3. Experimental procedure

In the present study, a set of experiments was performed for various operational parameters: a temperature range from 500 to 700 °C, a pressure 3 bar, steam-to-methane ratio from 1 to 4, a residence time (residence time i.e. catalyst pellet weight divided by the feed rate of the methane expressed as $\text{kg}_{\text{cat}} \cdot \text{s/mol}$ shows mass of catalyst bed per mole flow-rate on methane inlet) from 1 to 8 $\text{kg}_{\text{cat}} \cdot \text{s/mol}$. The residence time was varied due to a change in the flow rate of the reaction mixture.

Before the experiment, the electrical furnace heats the catalyst to the predetermined temperature. When the catalyst is heated, the reaction space of the methane reformer is filled with nitrogen. Heating of the catalyst takes approximately 15–17 min. The fresh catalysts are used for each experiment. During this period, the temperature in the reformer stabilizes throughout the reaction space. In parallel with the heating of the catalyst, steam generation is regulated by means of flowmeters. The temperature of the steam/methane reaction mixture is regulated in the heat exchanger to a temperature of the catalyst in the reformer.

At the initial time, the system of steam generation is not connected to the reformer. After the reformer becomes warmed up and the steam generation is regulated, the steam generation system is connected to the reformer. With the help of an electric heater, the temperature in the reaction space during the experiments is regulated. Each experiment has a duration of 10–15 min. Fresh catalyst is used for each experiment. During the experiment, the composition of the reformed products and the heat consumption for the reforming reaction are fixed every minute.

3. Results and discussion

3.1. Catalyst loading

The size and shape of catalyst particles play a big role in the SMR process because a lower dimensional model of steam methane reforming showed that majority of the reaction happens in a region 2–4 mm of the radius from the particle surface [41]. On the other hand, the weight of the catalyst affects the residence time [12,37]. Filling with various shapes of catalysts gives a different packed bed surface and weight of the packed bed in the reformer. Fig. 4 shows a comparison of loading characteristics of various packed beds filled with $\text{Ni-}\alpha\text{Al}_2\text{O}_3$ catalysts of various shapes.

As can be seen from Fig. 4 the packed bed filled with cylinders has a maximum weight and a minimum bed surface. While the packed bed filled with the spheres with 7-holes has a minimum weight and a maximum surface of the packed bed. In addition, the number of catalyst particles of the packed beds filled with cylinders, Raschig rings, and 7-holes cylinders is 256 pieces for all cases because these particles have

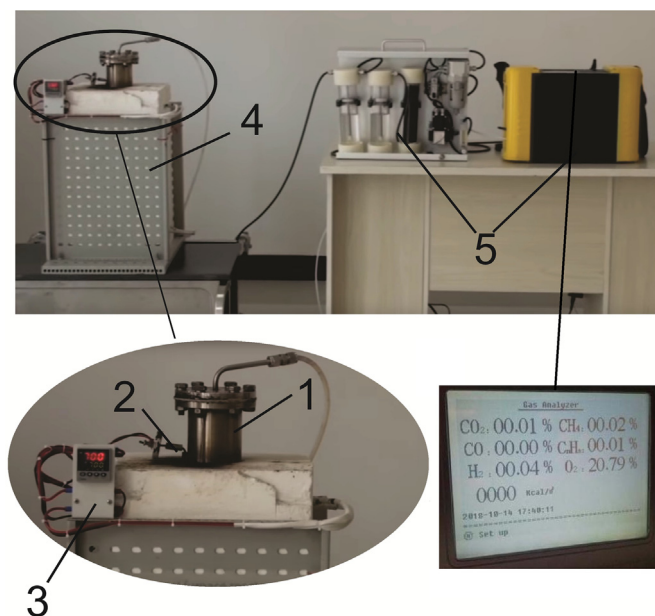


Fig. 2. Photograph of experimental setup.

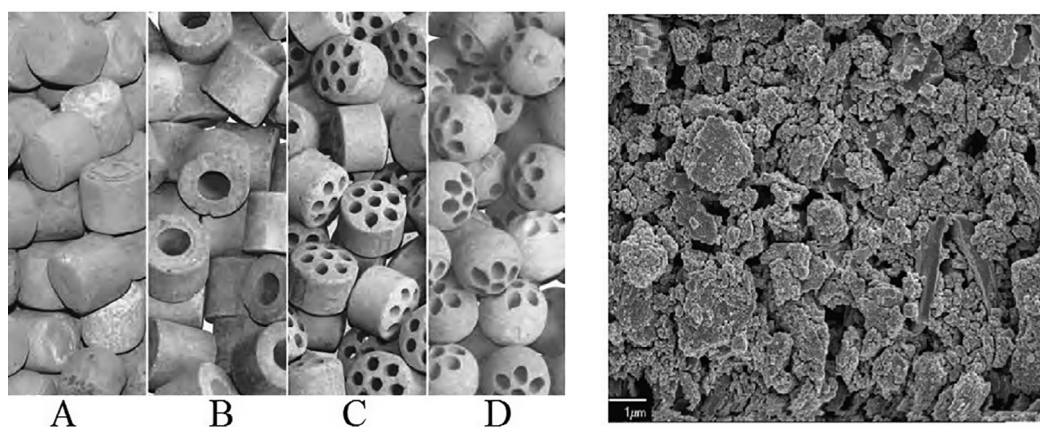


Fig. 3. Left side are photographs of the packed beds filled by the catalysts with different shapes: A – a cylinder; B – a Raschig ring; C – a convex cylinder with 7 internal holes; D – a sphere with 7 internal holes. Right side is SEM image for Ni-Al₂O₃ catalyst.

Table 1
The geometrical characteristics of catalyst.

Catalyst shape	Diameter, mm	Thickness, mm	Holes, number/diameter	Void fraction of packed bed
A	25	25	–	0.35
B	25	25	1 × 13	0.48
C	25	25	7 × 5	0.52
D	25	–	7 × 5	0.62

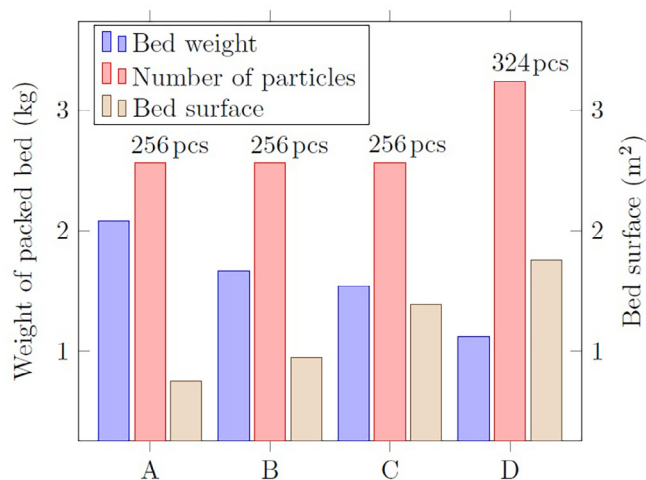


Fig. 4. Characteristics of catalyst loading.

the same end-to-end dimensions (25 mm × 25 mm). On the other hand, the number of particles of a packed bed filled with 7-holes spheres is 324 pieces despite the fact that the 7-holes spheres have the same end-to-end dimensions (25 mm × 25 mm).

3.2. Methane conversion

To analyze the effectiveness of the steam methane reforming process, the conversion of methane is defined as follows:

– methane conversion:

$$X_{CH_4} = \frac{CH_{4(in)} - CH_{4(out)}}{CH_{4(in)}} \quad (3)$$

At the first stage of the study, the methane conversion was determined for each catalyst bed at a constant mass flow rate of the

reaction mixture. The initial steam-to-methane ratio was CH₄:H₂O = 1:1. The mass flow rate of the reaction mixture is 15 g/s, while the mass flow rate of methane is 7.05 g/s. This 7.05 g/s of methane is equivalent to 0.44 mol/s molar flow rate of methane. All experiments were performed at a pressure of 3 bar. CH₄ conversion for various catalyst beds as a function of temperature is presented in Fig. 5.

Fig. 5 shows that CH₄ conversion depends on temperature and catalyst shapes. The minimum CH₄ conversion is achieved in a packed bed with cylinders, while the maximum CH₄ conversion takes place in a packed bed filled with -holes spheres. The discrepancy between the results for the cylinders and the 7-holes spheres reaches 23%. Such results can be explained by the fact that the volume of the catalyst particles in the packed bed filled with the 7-holes spheres is used more efficiently because of the larger surface of catalyst bed [41]. Herewith, in the packed bed filled with the cylinders, an inner part of the volume of the catalyst particles remains unused in the chemical reaction.

It is known that CH₄ conversion also depends on the duration of contact between the reaction mixture and the catalyst. This duration is called the contact time or the residence time. The contact time for the reaction mixture can be changed by changing the depth of the packed bed or changing the mass flow rate of the reaction mixture. Fig. 6 shows the variation of the methane conversion versus the mass flow rate of the reaction mixture at CH₄:H₂O = 1:1, p = 3 bar. The dashed and solid lines in Fig. 6 is an approximation curve of the experimental results.

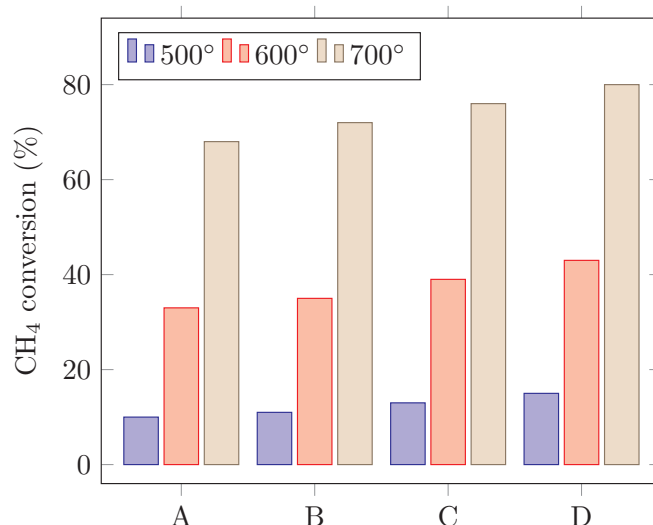


Fig. 5. CH₄ conversion vs. temperature at CH₄:H₂O = 1:1, $\dot{m}_{rm}$ = 15 g/s, p = 3 bar.

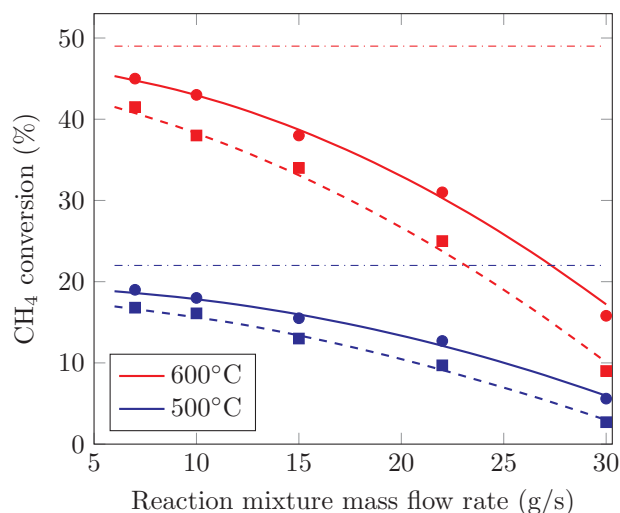


Fig. 6. CH₄ conversion vs. mass flow rate of reaction mixture at CH₄:H₂O = 1:1, $p = 3$ bar: square – cylinders; circle – 7-holes spheres. Solid and dashed lines – approximating curve for results that obtained when cylinders and 7-holes spheres is used; Dash-dotted line – expected thermodynamic value of CH₄ conversion.

Fig. 6 also presents the expected thermodynamic value of methane conversion. The thermodynamics equilibrium value was determined by Gibbs free energy minimization method. The algorithm of thermodynamic analysis by Gibbs free energy minimization method was discussed in author's previous articles [42,8] as well as in other authors articles [43]. The thermodynamic equilibrium values employing the Gibbs free energy minimization method were calculated using Aspen-HYSYS software, which is a powerful process simulator for modeling steady-state and unsteady-state process systems.

It can be seen from Fig. 6 that the methane conversion decreases with increasing mass flow rate of the reaction mixture. This is due to the fact that the residence time of the reaction mixture in the packed bed is inversely proportional to the mass flow rate of the reaction mixture, that is, the higher the mass flow rate, the lower the contact time of the reaction mixture and the catalyst. Moreover, Fig. 6 depicts that CH₄ conversion in a packed bed filled with 7-holes spheres is higher than in a packed bed filled with cylinders. This can be explained by the fact that the volume of the 7-holes spherical particles is used more efficiently than the volume of cylinders.

For more detailed analysis, two parameters are introduced: residence time for weight and residence time for surface. First (residence time for weight W_{cat}/F_{CH_4}) is calculated as catalyst pellet weight divided by the feed rate of the methane and expressed as kg_{cat}/mol. The weight residence time shows a mass of catalyst bed per mole flow-rate on methane inlet. Second (residence time for surface S_{cat}/F_{CH_4}) is calculated as a geometrical surface of a packed bed divided by the feed rate of the methane and expressed as m²_{cat}/mol. Fig. 7 depicts the methane conversion as a function of the residence time of CH₄ (weight or surface of a packed bed per mole of inlet CH₄) for steam reforming at a steam-to-methane ratio of 1 and different reforming temperatures.

As shown in Fig. 7, the experimental methane conversion in a packed bed filled with 7-holes spheres is increasing faster than CH₄ conversion in a packed bed filled with cylindrical particles for weight residence time. For example, for the weight residence time of 2 kg_{cat}/mol, CH₄ conversion obtained for the 7-holes spheres is two times as large than CH₄ conversion obtained for the cylinders. In this case, with a further increase in weight residence time, the discrepancy between CH₄ conversions decreases. At the weight residence time above 8 kg_{cat}/mol, the methane conversion for the 7-holes spheres and for the cylinders approaches each other. This can be explained by the fact that with a high weight residence time, the steam methane reforming

processes reaches a near-equilibrium condition. It is known that the catalyst shape does not affect the equilibrium composition of the steam methane reforming reaction.

The methane conversion versus surface residence time is shown in the right side of Fig. 7. The characteristic curve of a dependence of the CH₄ conversion on the surface residence time for the 7-holes spheres and the cylinders almost coincide in the range from 0 to 6 m²_{cat}/mol. Comparing the left and right parts of Fig. 7, it can be concluded that CH₄ conversion on the catalyst particles, the end-to-end dimensions of which is about 25 mm × 25 mm, is the limiting factor for achieving a given CH₄ conversion is primarily the surface of a packed bed. This is due to the fact that the reaction majority of the reaction occurs in a region 10–15% of the radius from the particle surface [41].

To understand the effect of particle shape on the CH₄ conversion at the various steam-to-methane ratio, a set of experiments was made. The experiments are made for constant weight and surface residence time. The results of these experiments are plotted in Fig. 8. Constant weight residence time is of about 5 kg_{cat}/mol and constant surface residence time is of about 3 m²_{cat}/mol. The residence times were changed by changing the mass flow rate of the inlet reaction mixture. As can be seen from Fig. 8 the methane conversion increases with increasing steam-to-methane ratio. The nature of the curves shows a good agreement with other results [37]. Fig. 8 confirms the assumption made above that, with the same surface residence time, the CH₄ conversion for different catalyst shapes is approximately equal.

To understand the effect of the size of the catalyst particle on the methane conversion, the set of experiments was made. The various catalyst particles were tested for the steam methane reforming process. Two catalyst shapes were used for these experiments: simple cylinders and Raschig rings. The end-to-end dimensions were 5 mm × 5 mm, 10 mm × 10 mm, 15 mm × 15 mm, 20 mm × 20 mm, and 25 mm × 25 mm (main diameter × height). The Raschig rings had an inner hole with the diameter of 1/2 main diameter. Fig. 9 shows the dependence of CH₄ conversion versus a size of the catalyst particle. The experiments were performed for a constant weight residence time (Fig. 9a) and for a constant weight residence time (Fig. 9b). The commercial Ni/α-Al₂O₃ catalyst is also used for these experiments. The solid lines are results that obtained for the Raschig ring, the dashed lines are for the simple cylinders. The filled points are the experimental results, the lines are the approximating curves. Fig. 9b does not have the filled points because they have approximately same coordinates for the simple cylinders and for the Raschig rings.

Fig. 9a depicts that with a decrease in the particle size of the catalyst, the degree of methane conversion increases (at a constant weight residence time). In addition, the smaller particle size gives the smaller the discrepancy between the CH₄ conversion. For example, the methane conversion at 600 °C for a simple cylinder with a size of 25 mm is 17% less than the methane conversion for a Raschig ring. At the same time, the methane conversion at 600 °C for a simple cylinder with a size of 5 mm is only 2.7% less than the methane conversion for the Raschig ring.

In contrast, Fig. 9b shows that the size of catalyst particles does not have an observable effect on the CH₄ conversion at a constant surface residence time about 3 m²/mol. The CH₄ conversions were roughly the same for 500 °C and 600 °C, and for all sizes. Based on these results, it can be argued that the SMR process occurs mainly in the area near the catalyst surface and a decrease in the particle size of the catalyst increases the efficiency of using its volume. These results are a good correlation with the results that obtained by Pedernera et al. [41]. Moreover, with a further decrease in the particle size of the catalyst, the methane conversion will increase and an equilibrium CH₄ conversion will be achieved at a lower weight residence time. For particle sizes less than 3 mm, Xu and Froment [12] recommends using only weight residence time, since the entire volume of the catalyst particle is involved in the SMR reaction.

Conspicuously from Fig. 9 the catalytic activity of the surface unit of

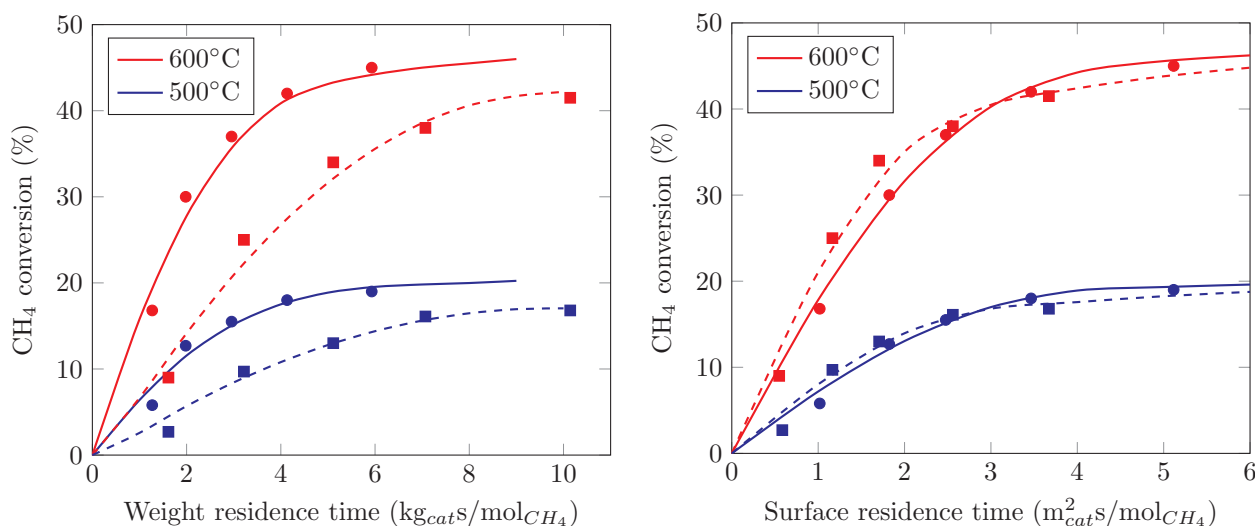


Fig. 7. CH₄ conversion vs. residence time at CH₄:H₂O = 1:1, p = 3 bar: left figure – weight residence time; right figure – surface residence time; square – cylinders; circle – 7-holes spheres.

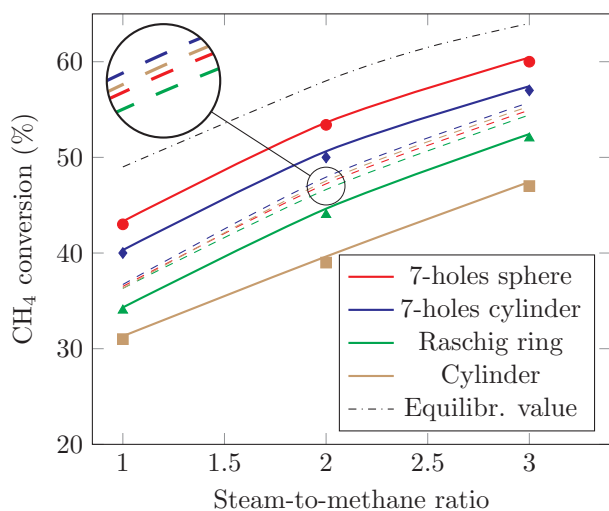


Fig. 8. CH₄ conversion vs. steam-to-methane ratio at 600°C and 3 bar. Solid line – weight residence time, dashed line – surface residence time, dash-dotted line – expected thermodynamic value of CH₄ conversion.

these catalysts are virtually the same. Therefore, in order to improve the performance of the steam methane reforming reactor, it was rational to increase the geometric surface of the packed bed. An increase in the geometric surface of the packed bed will allow maintaining the methane conversion with a decrease in the weight of the catalyst in the reformer. However, without changing other characteristics of the granular layer, this causes an undesirable increase in pressure drop. However, a change in the geometric surface due to compaction of the packed bed may cause an undesirable increase in a pressure drop.

3.3. Pressure drop

As can be seen from Fig. 3, the steam methane reformer is filled by a densely packed bed. The catalyst shape has a noticeable effect on the pressure drop in the reformer. The pressure difference between the bottom and top of the reformer is the “pressure drop (ΔP)”. The pressure drops were measured for various mass-flow rates in the temperature range from 500 to 600°.

Fig. 10 presents the results of the pressure drops measuring during the steam methane reforming process at 600°C, H₂O:CH₄ = 1:1,

p = 3 bar. The packed beds have 600 mm in a length and 100 mm in a diameter. The filled dots are the experimental results, the solid lines are the approximating curves. Each mass flow rate was tested for 5 times. The filled points in Fig. 10 are the arithmetic mean value of the pressure drops. It can be observed that the pressure drops increase with increasing mass flow rate (and as a result, inlet velocity). Moreover, the increase in pressure has a characteristic quadratic dependence. This observation is a good correlation with experimental data [44] and CFD-modeling results [29,45].

From the point of view of the classical representation of fluid dynamics, the pressure drops in the packed bed can be considered as the local hydraulic losses, which are caused by changes in the shape and size of the channel, which is deforming the flow. It can be observed that from Fig. 10 the pressure drop in a packed bed is proportional to the square of the velocity. Therefore, the pressure drop can be written as:

$$\frac{\Delta P}{L} = C_d \frac{\rho u^2}{2}, \quad (4)$$

where L – length of a packed bed, m; C_d – drag coefficient of a packed bed; ρ – average density of reaction mixture in a packed bed, kg/m³; u – reaction mixture velocity, m/s.

In Eq. (4), the drag coefficient (C_d) includes both hydrodynamic drag forces on particles and the wall of reformer tube. The variations of the drag coefficient for 1 m of the packed bed versus the velocity for the different packed beds are shown in Fig. 11.

Fig. 11 shows that the drag coefficients are varying substantially when the velocity varies, especially in the range up to 1 m/s. Usually, the drag coefficient for the solid non-porous bodied depends weakly on the velocity. For the dependences of the drag coefficients versus velocity, it can be observed that the drag coefficients are heavily depended on velocity. This effect of velocity on the coefficient can be explained by the fact that with increasing velocity, the amount of gas that passes through the holes in catalyst increases. At the same time, when the gas velocity is low, the fluid energy is not enough to percolate through the porous structure of a Ni- α -Al₂O₃ catalyst.

In practice, the values of drag coefficients that obtained in Fig. 11 is difficult to use, because these coefficients are taking into account only the length of the packed beds. For engineering calculations, it is convenient to use the drag coefficients for 1 m³ of the packed bed. Table 2 shows the volume drag coefficient. Therefore, the pressure drop of 1 m³ of the packed beds can be calculated as:

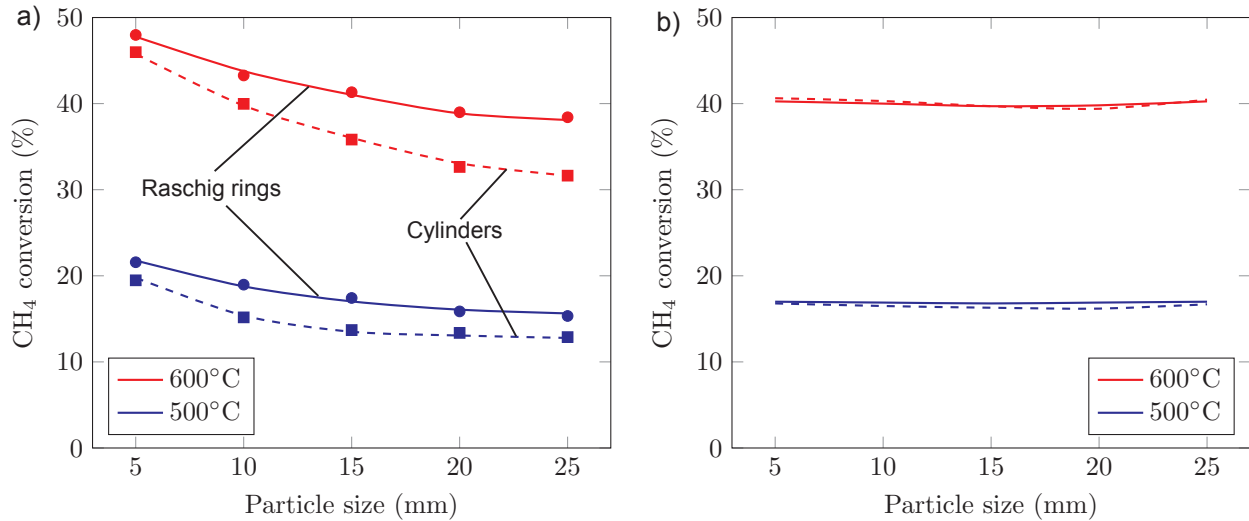


Fig. 9. CH₄ conversion vs. size of catalyst particle at CH₄:H₂O = 1:1 and 3 bar: a) weight residence time is constant and equal of about 5 kg_{cat}/mol; b) surface residence time is constant and equal of about 3 m²/mol.

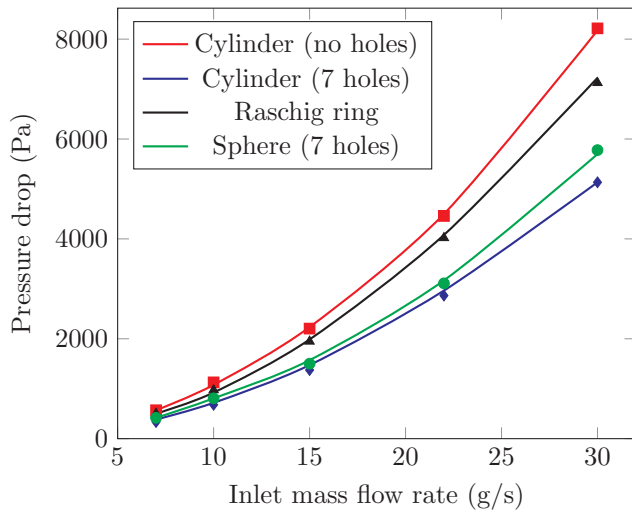


Fig. 10. Pressure drop vs. inlet mass flow rate for various catalyst shapes for the packed beds that presented in Table 1.

$$\frac{\Delta P}{V_{pb}} = C_d^V \frac{\rho u^2}{2}, \quad (5)$$

where V_{pb} – volume of a packed bed, m³; C_d^V – volume drag coefficient, 1/m³.

4. Conclusion

In this paper, the results of an experimental investigation of the steam methane reforming process are presented. The investigation is mainly focused on the study of the effect of the catalyst shapes on the methane conversion and pressure loss in the packed bed. The main conclusions can be summarized as follows:

- The shape of catalyst particle has a noticeable effect on the methane conversion. The discrepancy between the results for the cylinders and the 7-holes spheres reaches 23% at 600 °C and H₂O:CH₄ = 1:1. Such results can be explained by the fact that the volume of the catalyst particles in the packed bed filled with the 7-holes spheres is used more efficiently because of the larger surface of the catalyst bed.
- The catalytic activity of the surface unit of all investigated catalyst

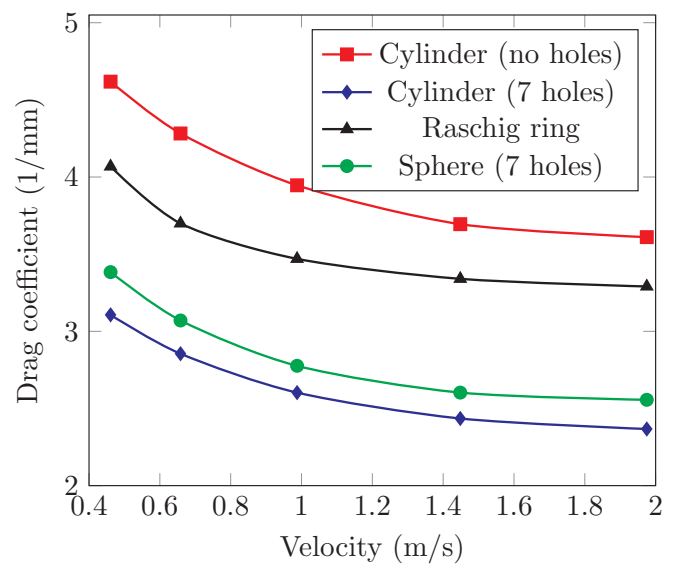


Fig. 11. Drag coefficient of 1 mm of a packed bed vs. reaction mixture velocity for the packed beds that presented in Table 1.

Table 2

The volume drag coefficient.

Velocity, m/s	Volume drag coefficient (C_d^V), 1/m ³			
	Simple cylinder	Raschig ring	7-Holes cylinder	7-Holes sphere
0.5	588289.9	518229.9	395758.7	431056.1
0.75	545505.2	467423.1	363670.1	408950.6
1.0	502720.5	447100.3	331581.6	353687.0
1.5	470631.9	426777.6	310189.2	331581.6
2.0	459935.7	406454.8	288796.9	320528.9

shapes is virtually the same. Therefore, in order to improve the performance of the steam methane reforming reactor, it is rational to increase the geometric surface of the packed bed. Reducing the size of the catalyst particles increases the methane conversion at a constant weight residence time.

- A characteristic quadratic dependence of a pressure drop versus velocity $\Delta P = f(u^2)$ is observed for all catalyst types.

- The drag coefficients of unit length are varying substantially when the velocity varies, especially in the velocity range up to 1-m/s.
- The volume drag coefficient of an unit volume is determined for engineering calculations of the packed bed filled with Ni- α -Al₂O₃ catalysts of various shapes.

Acknowledgements

This work was carried out with financial support from the grants of the President of the Russian Federation (Council on grants of the President of the Russian Federation, Grant/Award Number: MK-447.2019.8).

Conflict of interests

None.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.enconman.2019.01.103>.

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